Solution: Count of Subset Sum

Problem Statement

Given a set of positive numbers, find the total number of subsets whose sum is equal to a given number 'S'.

Example 1:

```
Input: {1, 1, 2, 3}, S=4
Output: 3
The given set has '3' subsets whose sum is '4': {1, 1, 2}, {1, 3}, {1,
3}
Note that we have two similar sets {1, 3}, because we have two '1' in
our input.
```

Example 2:

```
Input: {1, 2, 7, 1, 5}, S=9
Output: 3
The given set has '3' subsets whose sum is '9': {2, 7}, {1, 7, 1}, {1,
2, 1, 5}
```

Basic Solution

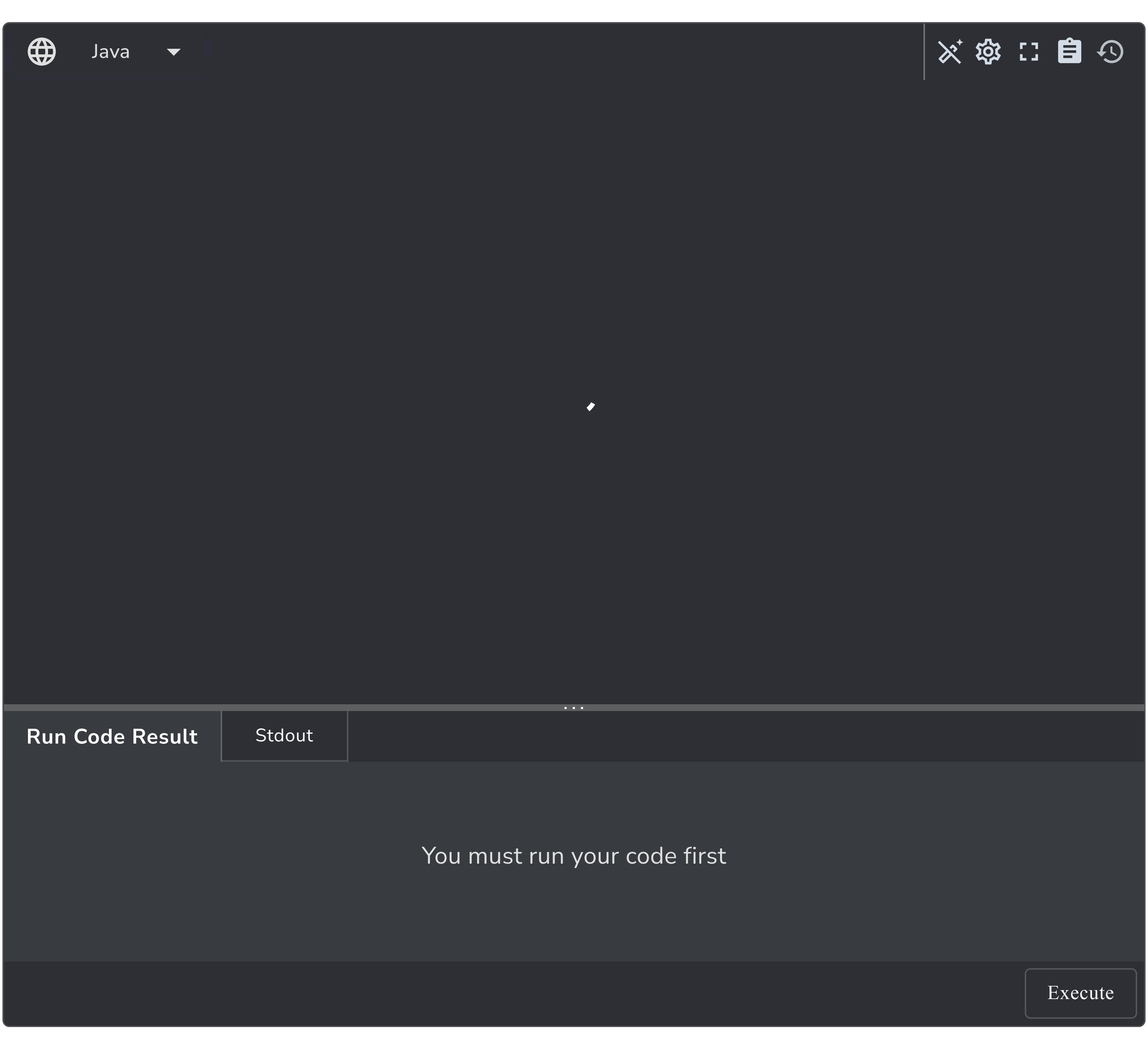
This problem follows the **0/1 Knapsack pattern** and is quite similar to `"Subset Sum"`. The only difference in this problem is that we need to count the number of subsets, whereas in the `"Subset Sum"` we only wanted to know if there exists a subset with the given sum.

A basic brute-force solution could be to try all subsets of the given numbers to count the subsets that have a sum equal to 'S'. So our brute-force algorithm will look like:

```
for each number 'i'
  create a new set which includes number 'i' if it does not exceed 'S', and recursively
      process the remaining numbers and sum
  create a new set without number 'i', and recursively process the remaining numbers
return the count of subsets who has a sum equal to 'S'
```

Code

Here is the code for the brute-force solution:



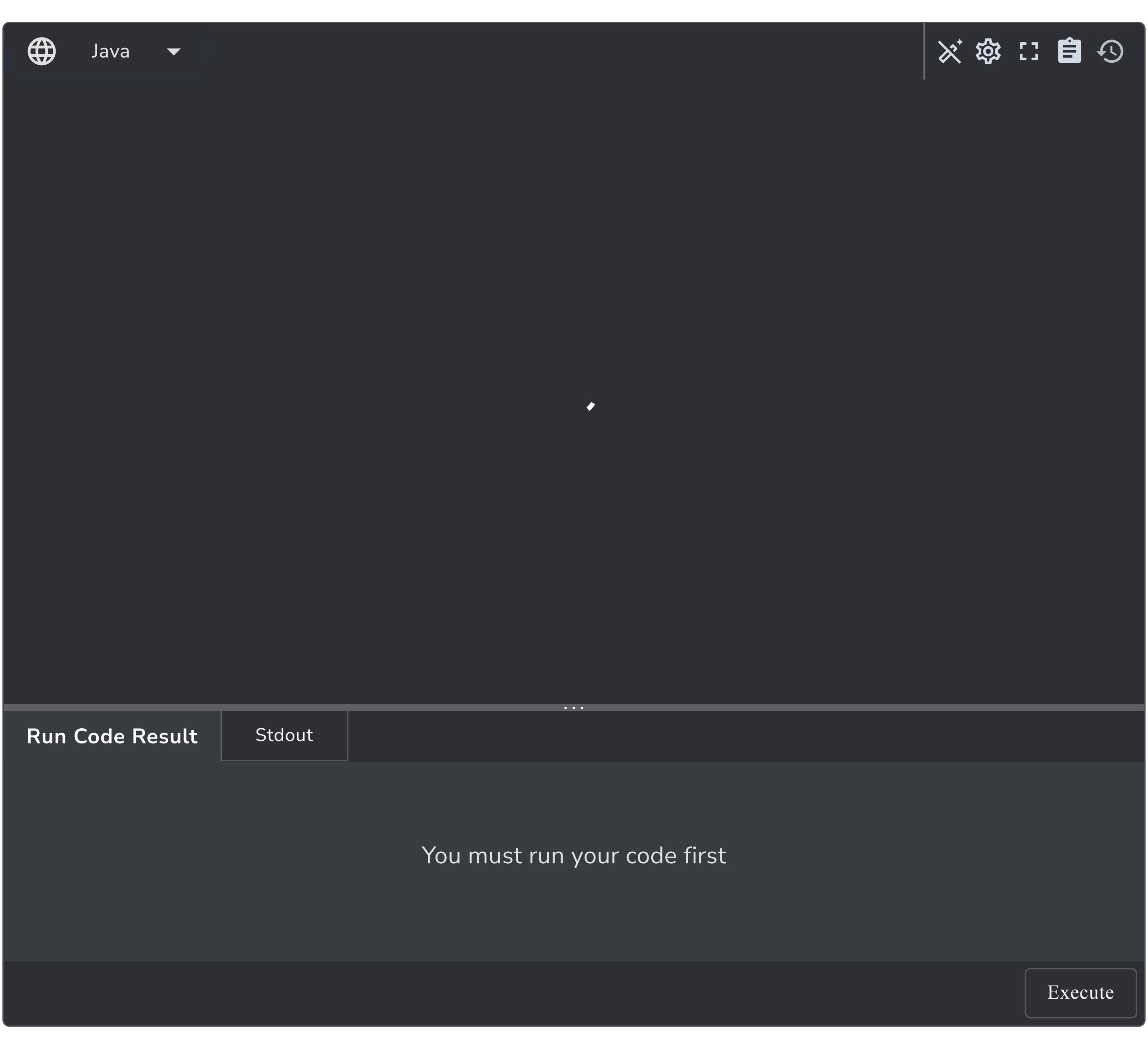
The time complexity of the above algorithm is exponential $O(2^n)$, where 'n' represents the total number. The space complexity is $O(n)$, this memory is used to store the recursion stack.

Top-down Dynamic Programming with Memoization

We can use memoization to overcome the overlapping sub-problems. We will be using a two-dimensional array to store the results of solved sub-problems. As mentioned above, we need to store results for every subset and for every possible sum.

Code

Here is the code:



Bottom-up Dynamic Programming

We will try to find if we can make all possible sums with every subset to populate the array `db[TotalNumbers][S+1]`.

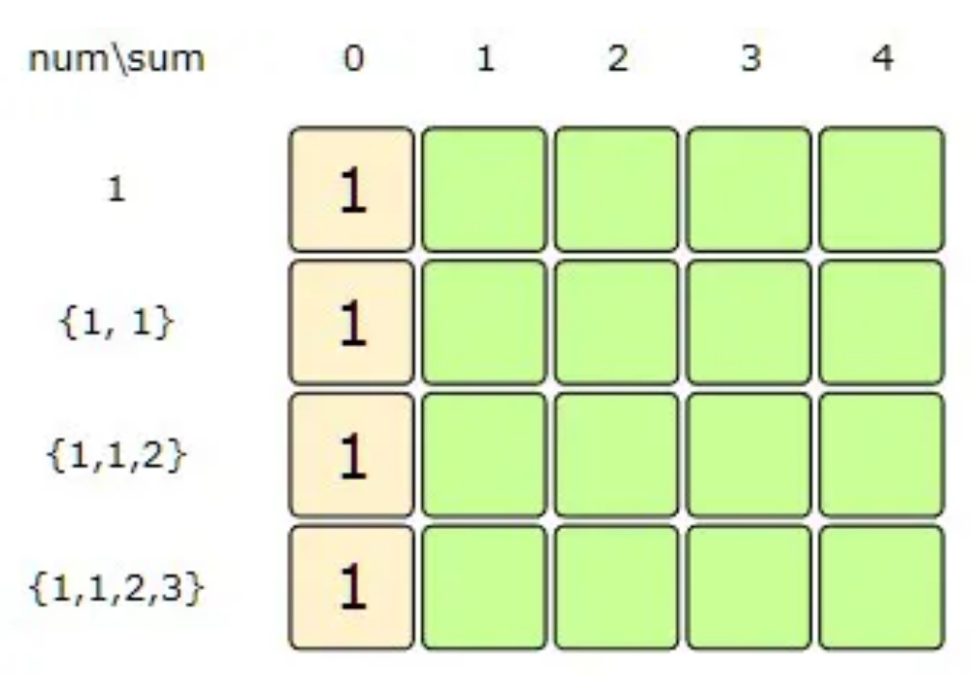
So, at every step we have two options:

1. Exclude the number. Count all the subsets without the given number up to the given sum => `dp[index-1][sum]`
2. Include the number if its value is not more than the 'sum'. In this case, we will count all the subsets to get the remaining sum => `dp[index-1][sum-num[index]]`

To find the total sets, we will add both of the above two values:

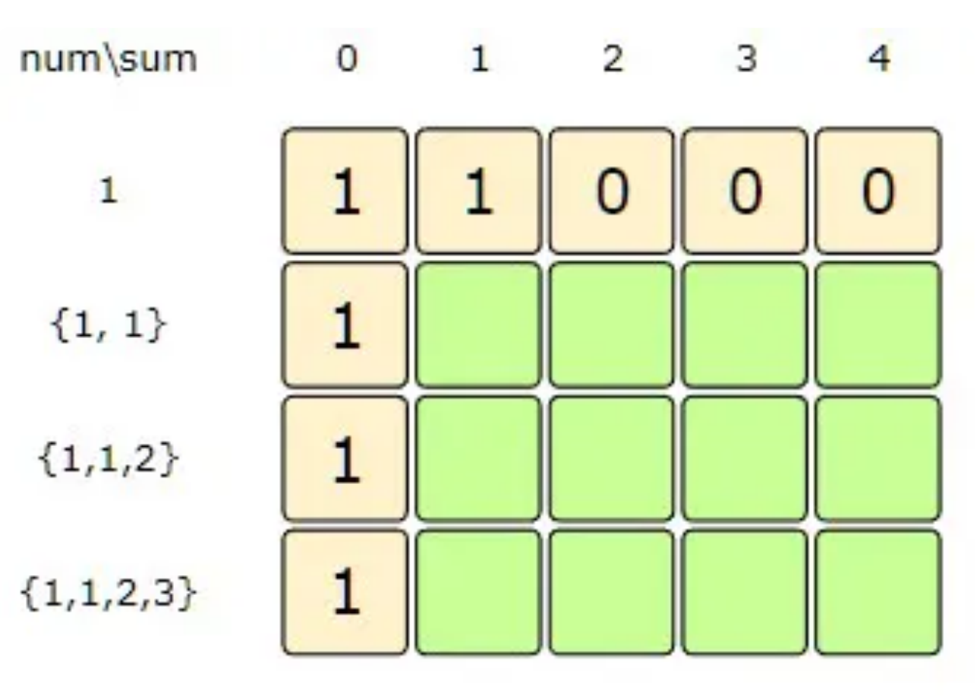
```
dp[index][sum] = dp[index-1][sum] + dp[index-1][sum-num[index]])
```

Let's start with our base case of size zero:



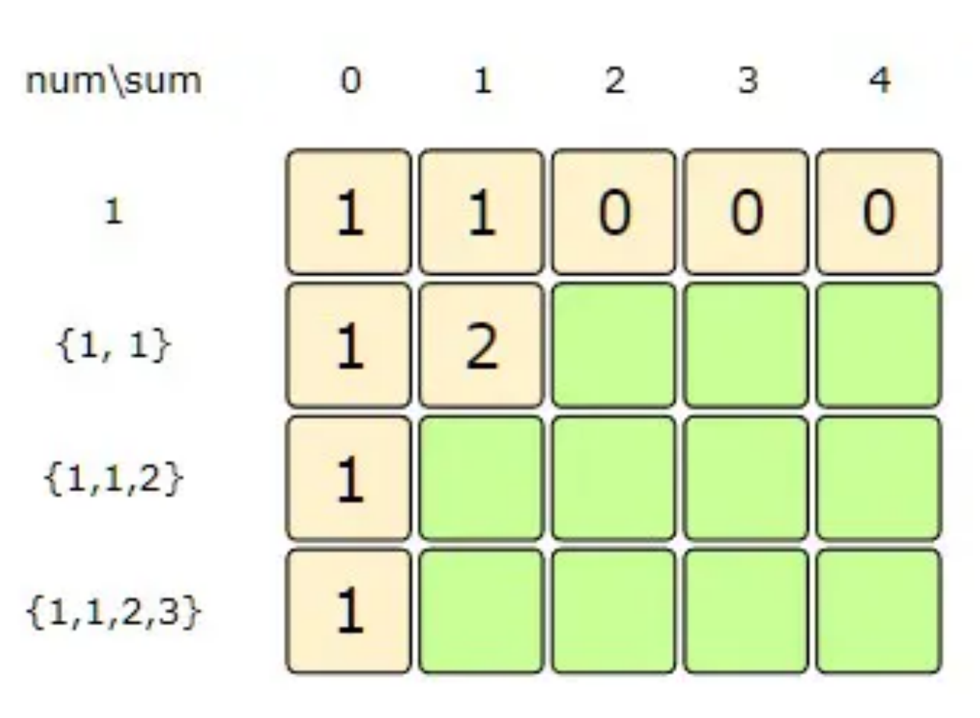
num\sum	0	1	2	3	4
1	1				
{1, 1}	1				
{1,1,2}	1				
{1,1,2,3}	1				

'0' sum can always be found through an empty set



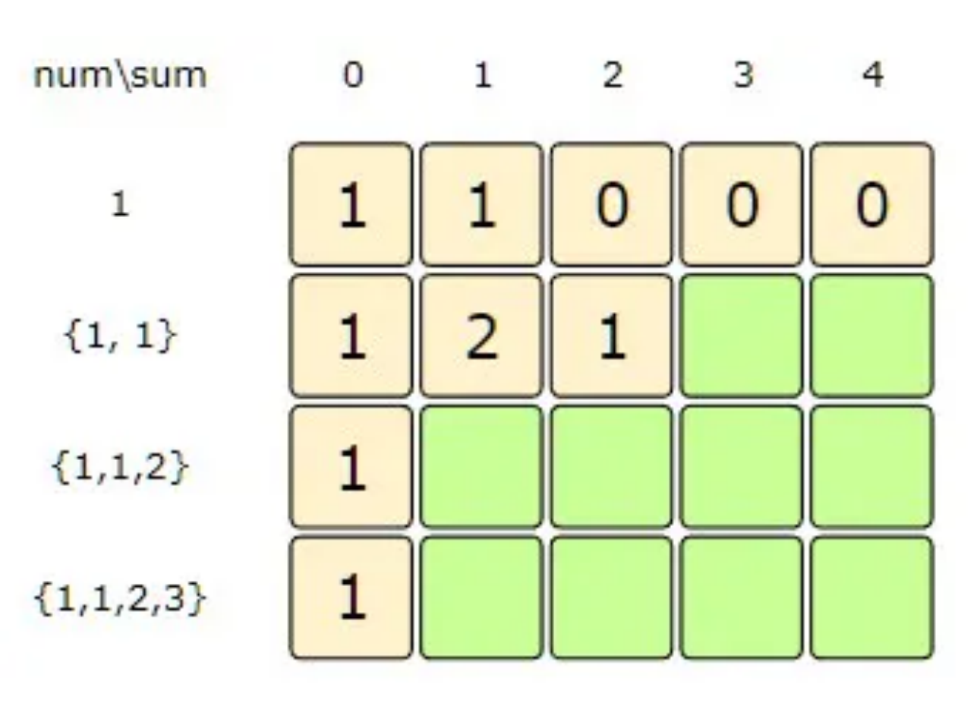
num\sum	0	1	2	3	4
1	1	1	0	0	0
{1, 1}	1				
{1,1,2}	1				
{1,1,2,3}	1				

With only one number, we can form a subset only when the required sum is equal to the number



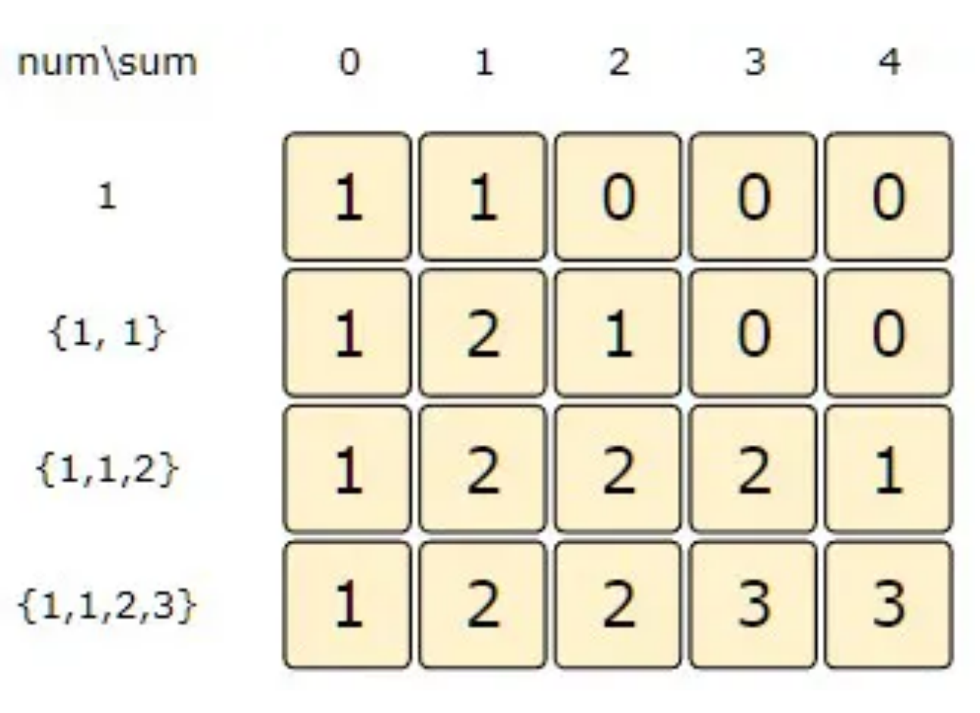
num\sum	0	1	2	3	4
1	1	1	0	0	0
{1, 1}	1	2			
{1,1,2}	1				
{1,1,2,3}	1				

sum: 1, index:1=> (dp[index-1][sum] + dp[index-1][sum - 1])



num\sum	0	1	2	3	4
1	1	1	0	0	0
{1, 1}	1	2	1		
{1,1,2}	1				
{1,1,2,3}	1				

sum: 2, index:1=> (dp[index-1][sum] + dp[index-1][sum - 1])

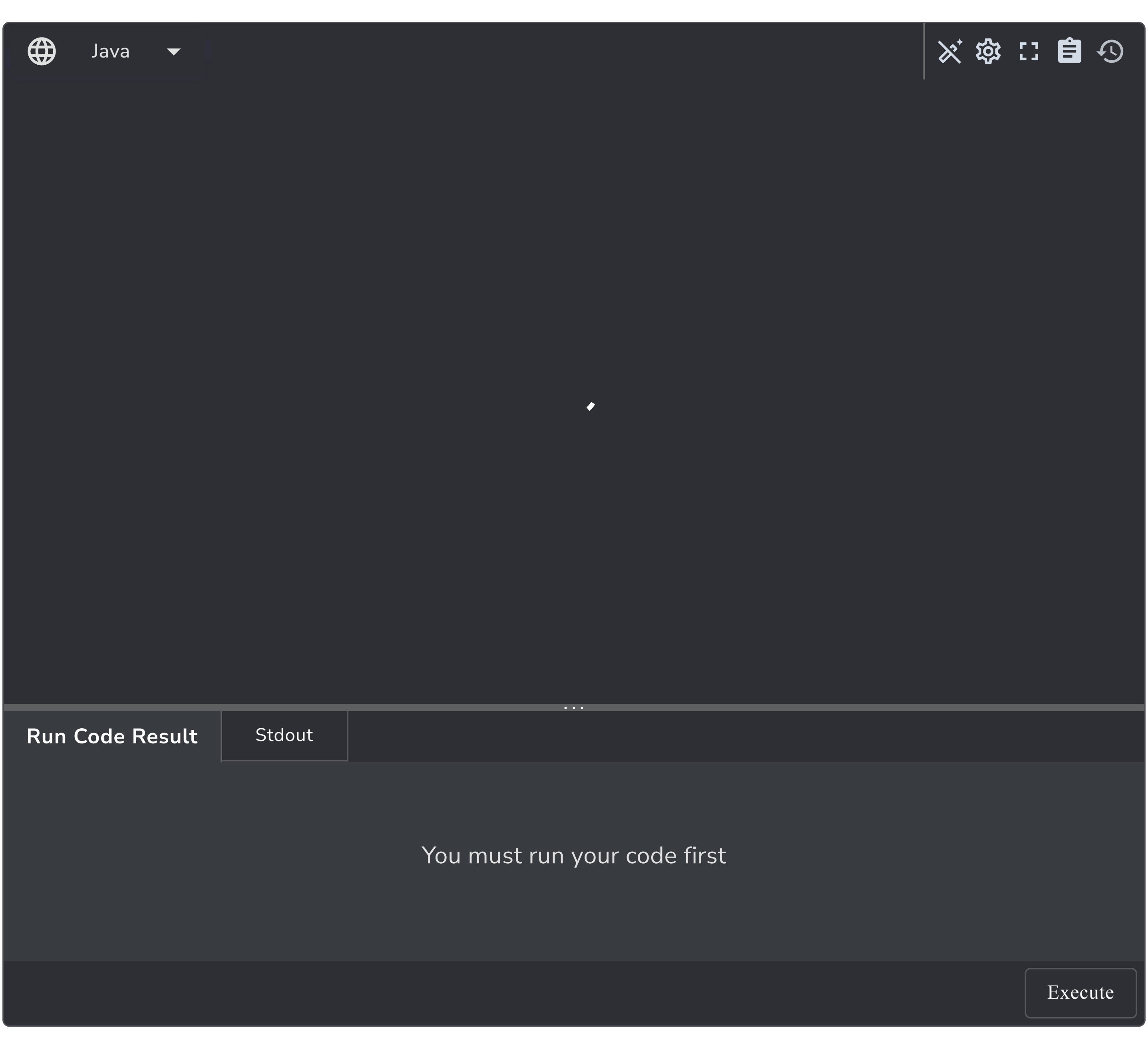


num\sum	0	1	2	3	4
1	1	1	0	0	0
{1, 1}	1	2	1	0	0
{1,1,2}	1	2	2	2	1
{1,1,2,3}	1	2	2	3	3

sum: 4, index:3=> (dp[index-1][sum] + dp[index-1][sum - 3])

Code

Here is the code for our bottom-up dynamic programming approach:

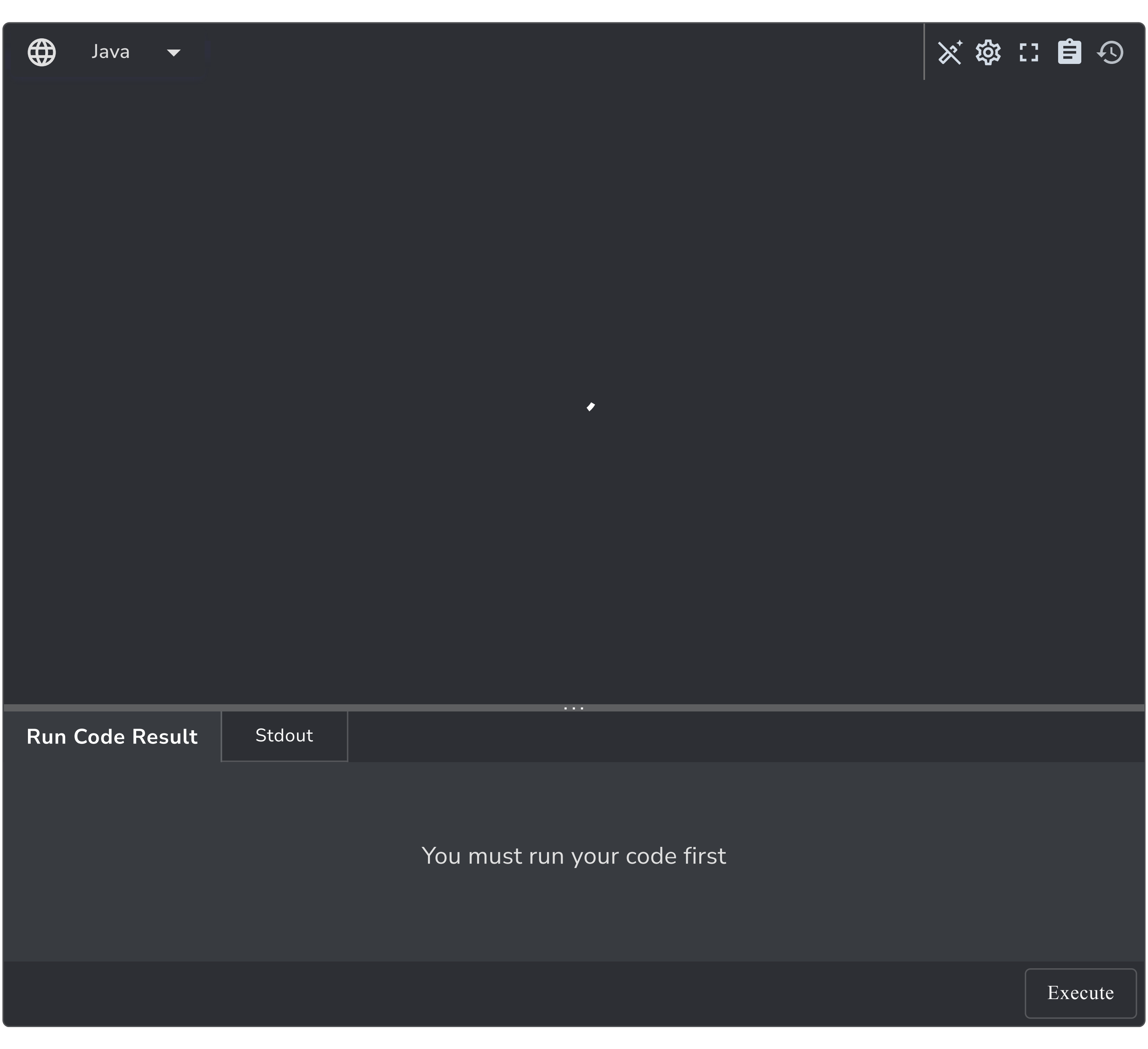


The above solution has time and space complexity of $O(N * S)$, where 'N' represents total numbers and 'S' is the desired sum.

Challenge

Can we further improve our bottom-up DP solution? Can you find an algorithm that has $O(S)$ space complexity?

```
Similar to the space optimized solution for "0/1 Knapsack."
```



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